

1. Two particles, P and Q , are moving towards each other in opposite directions along the same straight line when they collide directly. Immediately before the collision the speed of Q is $2u$. The mass of Q is $3m$ and the magnitude of the impulse exerted by P on Q in the collision is $4mu$.

Find

- (a) the speed of Q immediately after the collision,

(3)

- (b) the direction of motion of Q immediately after the collision.

(1)

2. A motorbike is moving with constant acceleration along a straight horizontal road.

The motorbike passes a point P and 10 seconds later passes a point Q .

The speed of the motorbike as it passes Q is 28 m s^{-1}

Given that $PQ = 220 \text{ m}$,

(a) find the acceleration of the motorbike,

(3)

(b) find the distance travelled by the motorbike during the fifth second after passing P

(4)

3. A tractor of mass 6 tonnes is dragging a large block of mass 2 tonnes along rough horizontal ground. The cable connecting the tractor to the block is horizontal and parallel to the direction of motion.

The cable is modelled as being light and inextensible.

The driving force of the tractor is 7400 N and the resistance to the motion of the tractor is 200 N. The resistance to the motion of the block is R newtons, where R is a constant.

Given that the tension in the cable is 6000 N and the tractor is accelerating,

(a) find the value of R .

(6)

(b) State how you have used the fact that the cable is modelled as being inextensible.

(1)

4.

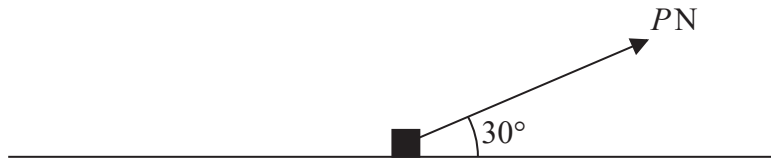


Figure 1

A small block of mass 5 kg lies at rest on a rough horizontal plane.

The coefficient of friction between the block and the plane is $\frac{3}{7}$

A force of magnitude P newtons is applied to the block in a direction which makes an angle of 30° with the plane, as shown in Figure 1.

The block is modelled as a particle.

Given that $P = 14$

(a) find the magnitude of the frictional force exerted on the block by the plane and describe what happens to the block, justifying your answer.

(6)

The value of P is now changed so that the block is on the point of slipping along the plane.

(b) Find the value of P

(6)

5.

**Figure 2**

A uniform rod AB has length 5 m and mass 5 kg. The rod rests in equilibrium in a horizontal position on two supports C and D , where $AC = 1$ m and $DB = 2$ m, as shown in Figure 2.

A particle of mass 10 kg is placed on the rod at A and a particle of mass M kg is placed on the rod at B . The rod remains horizontal and in equilibrium.

- (a) Find, in terms of M , the magnitude of the reaction on the rod at C . (3)
- (b) Find, in terms of M , the magnitude of the reaction on the rod at D . (3)
- (c) Hence, or otherwise, find the range of possible values of M . (3)
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6. A particle P is moving with constant acceleration.

At time $t = 1$ second, P has velocity $(-\mathbf{i} + 4\mathbf{j})\text{ m s}^{-1}$

At time $t = 4$ seconds, P has velocity $(5\mathbf{i} - 8\mathbf{j})\text{ m s}^{-1}$

Find the speed of P at time $t = 3.5$ seconds.

(6)

7. Two small children, Ajaz and Beth, are running a 100 m race along a straight horizontal track.

They both start from rest, leaving the start line at the same time.

Ajaz accelerates at 0.8 m s^{-2} up to a speed of 4 m s^{-1} and then maintains this speed until he crosses the finish line.

Beth accelerates at 1 m s^{-2} for T seconds and then maintains a constant speed until she crosses the finish line.

Ajaz and Beth cross the finish line at the same time.

- (a) Sketch, on the same axes, a speed-time graph for each child, from the instant when they leave the start line to the instant when they cross the finish line.

(3)

- (b) Find the time taken by Ajaz to complete the race.

(4)

- (c) Find the value of T

(4)

- (d) Find the difference in the speeds of the two children as they cross the finish line.

(2)

8. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

Two boats, P and Q , are moving with constant velocities.

The velocity of P is $15\mathbf{i} \text{ m s}^{-1}$ and the velocity of Q is $(20\mathbf{i} - 20\mathbf{j}) \text{ m s}^{-1}$

- (a) Find the direction in which Q is travelling, giving your answer as a bearing.

(2)

The boats are modelled as particles.

At time $t = 0$, P is at the origin O and Q is at the point with position vector $200\mathbf{j} \text{ m}$.

At time t seconds, the position vector of P is $\mathbf{p} \text{ m}$ and the position vector of Q is $\mathbf{q} \text{ m}$.

- (b) Show that

$$\overrightarrow{PQ} = [5t\mathbf{i} + (200 - 20t)\mathbf{j}] \text{ m}$$

(5)

- (c) Find the bearing of P from Q when $t = 10$

(2)

- (d) Find the distance between P and Q when Q is north east of P

(5)

- (e) Find the times when P and Q are 200 m apart.

(3)
